Reg. No. | | | | | | | | | | | [16CSS01605]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

LINUX ADMINISTRATION

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions selecting atleast two from each part.

PART-A

1.	a.	Explain the architecture of Linux and discuss the characteristics of Linux operating system.	08
	b.	What are the default ports used for SMTP, DNS, FTP, DHCP, SSH and Differentiate between Absolute path and Relative path in linux.	06
2.	a.	Illustrate the booting process of the Linux operating system with the help of diagram.	05
	b.	Describe Linux file system hierarchy. Explain the term i-node. Explain i-node in brief with the help of Directory structure.	09
3.	a.	Exemplify the term "Process" in Linux operating system and Explain all the types of process. What are the various ID associated with the process.	06
	b.	Demonstrate the management of User and Group permissions through commands and list out the main roles and responsibilities of system administration.	08
4.	a.	Create disk partitioning in Linux system using f-disk utility commands. Describe all the file systems of Linux along with its features.	06
	b	With a neat diagram describe logical Volume management briefly and list the features of Logical Volume Management.	08

PART-B

5.	a.	Explain the term "apt" (Advance packaging tool) with the help of examples and list the benefits of package management in Linux.	07
	b.	Demonstrate the FTP server with the help of commands and write down the benefits of FTP server over telnet.	07
6.	a.	Give a short brief on Network file server. Write down the steps to create the NFS server in Linux system.	09
	b.	Write a short note on smb protocol (samba) and write down the steps to configure samba server.	05
7.	a.	Describe the Runlevel modes of Unix based operating system. How to manage the services of Linux and infrastructure services with the help of Systemd and demonstrate with the help of systemctl command.	09
	b.	With the help of Diagram, Explain Briefly how DNS will work. List out the advantages of Domain name server.	05
8.	a.	System administrators need to consider how strict a policy to implement for their server environment. How SELinux will help to provide more secure environment to server? Discuss the modes of SElinux with the help of examples.	08
	b.	Differentiate between the SSH protocol and telnet. What is the Default Port & Configuration File of ssh Server?	06

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